

BEVERLY, MA, USA

WMO: 725088

Lat: 42.584N

Lon: 70.918W

Elev: 33

StdP: 100.93

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 54733

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.6	-12.5	-24.8	0.4	-13.3	-22.1	0.5	-11.0	11.6	-4.1	10.8	-3.7	4.6	310	0.532

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	32.0	22.9	30.1	22.3	28.3	21.4	24.4	29.7	23.5	28.1	22.7	26.7	4.8	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.8	17.5	27.0	22.2	16.9	26.2	21.3	16.0	25.2	73.8	29.7	70.2	27.8	67.0	26.7	26.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.9	8.6	7.7	DB	-18.6	34.6	3.6	1.7	-21.2	35.9	-23.3	36.9	-25.3	37.9	-27.9	39.2	(4)
(5)			WB	-19.3	25.9	3.1	0.7	-21.5	26.4	-23.3	26.8	-25.0	27.2	-27.3	27.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.9	-2.5	-1.3	2.3	8.3	13.5	18.6	22.0	21.4	17.5	11.3	5.9	0.8	(6)	
	DBStd	9.62	5.92	5.12	5.00	4.53	4.26	4.05	3.20	3.19	3.93	4.32	4.68	5.06	(7)	
	HDD10.0	1509	389	319	244	84	11	0	0	0	1	36	138	287	(8)	
	HDD18.3	3416	647	551	497	303	161	45	6	8	61	223	372	542	(9)	
	CDD10.0	1467	1	1	6	34	120	258	373	353	226	75	16	3	(10)	
	CDD18.3	331	0	0	0	2	12	53	121	103	37	3	0	0	(11)	
	CDH23.3	2654	0	0	2	36	132	436	986	783	257	23	0	0	(12)	
	CDH26.7	803	0	0	0	10	39	131	322	231	69	1	0	0	(13)	
(14) Wind	WSAvg	3.5	4.0	4.0	4.2	4.0	3.5	3.2	3.0	2.8	2.9	3.4	3.6	3.8	(14)	
(15) Precipitation	PrecAvg	1221	101	88	114	103	94	98	86	92	101	114	103	117	(15)	
	PrecMax	1617	161	214	426	270	422	275	218	218	210	320	202	191	(16)	
	PrecMin	856	32	44	18	12	17	6	35	30	42	6	28	38	(17)	
	PrecStd	183	31	38	84	55	77	64	49	49	52	80	41	39	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.9	15.9	21.1	27.9	30.9	32.9	34.1	33.7	32.2	26.2	20.9	17.2	(19)	
		MCWB	13.7	11.9	12.6	16.7	19.3	23.5	24.4	24.3	22.9	18.8	16.2	14.0	(20)	
	2%	DB	11.5	11.3	16.0	22.3	27.1	30.1	31.9	31.2	28.5	23.0	17.7	13.3	(21)	
		MCWB	9.0	7.6	10.6	14.7	18.5	22.5	22.9	23.3	21.2	17.9	14.0	10.6	(22)	
	5%	DB	8.1	8.6	12.2	18.8	23.8	27.7	30.2	29.0	26.2	20.7	15.6	10.7	(23)	
		MCWB	5.6	4.9	7.5	12.6	17.0	21.0	22.5	21.9	20.0	16.9	11.8	8.1	(24)	
	10%	DB	5.5	6.0	9.3	16.0	21.1	25.9	28.1	27.5	24.0	18.0	13.1	8.4	(25)	
		MCWB	2.9	2.7	5.5	10.5	15.7	19.6	21.6	21.2	19.4	14.5	10.3	6.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.9	13.1	14.5	18.0	21.6	24.7	25.7	25.5	24.0	21.3	17.6	14.6	(27)	
		MCDB	15.4	14.6	18.7	24.2	27.9	30.8	31.8	31.5	28.6	23.8	19.7	16.3	(28)	
	2%	WB	9.4	8.4	11.4	15.8	19.9	23.4	24.5	24.3	22.6	18.9	15.1	11.5	(29)	
		MCDB	11.1	10.8	15.5	21.3	25.4	28.6	29.9	28.9	26.3	21.7	16.9	13.1	(30)	
	5%	WB	5.9	5.2	8.0	13.6	18.2	22.2	23.5	23.4	21.6	17.2	12.8	8.7	(31)	
		MCDB	7.8	8.1	11.2	18.0	22.2	26.4	28.2	27.4	24.7	20.0	14.5	10.0	(32)	
	10%	WB	3.2	3.2	5.9	10.9	16.4	20.9	22.6	22.5	20.5	15.5	10.8	6.5	(33)	
		MCDB	5.1	5.4	8.8	14.7	20.4	24.5	26.8	26.0	23.3	17.8	12.9	8.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.4	9.2	8.9	9.8	9.8	9.8	9.9	10.0	10.3	9.8	9.0	8.1	(35)	
		MCDBR	11.1	11.9	12.8	14.9	14.3	12.2	12.0	11.6	12.3	12.2	11.3	10.3	(36)	
	5% WB	MCWBR	9.7	9.0	8.6	8.6	7.8	6.4	5.3	5.4	6.3	7.6	8.6	9.2	(37)	
		MCWBR	10.9	11.3	12.0	13.6	12.5	11.3	10.2	9.8	9.8	10.8	9.4	10.0	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.289	0.298	0.320	0.364	0.400	0.445	0.464	0.429	0.373	0.332	0.308	0.292	(40)	
		taud	2.514	2.506	2.504	2.412	2.350	2.257	2.209	2.295	2.434	2.553	2.578	2.574	(41)	
	Ebn at Noon	Ebn at Noon	865	913	929	904	875	833	812	830	858	862	836	832	(42)	
		Edh at Noon	70	83	94	112	122	134	140	124	101	79	66	62	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.73	2.60	3.61	4.64	5.18	5.59	5.90	5.31	4.26	2.81	1.90	1.42	(44)	
	RadStd	RadStd	0.11	0.21	0.24	0.50	0.58	0.52	0.37	0.23	0.28	0.24	0.15	0.16	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	+0.78	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+70	(46)	
(47)	Regional (74 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+90	+38	(47)	

Nomenclature: See separate page